

1. 设 $A = \begin{pmatrix} 1 & 1 \\ 2 & 5 \end{pmatrix}$, $x = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$, 试求 $\|A\|_1$, $\|A\|_2$, $\|A\|_\infty$, $\|Ax\|_1$, $\|Ax\|_2$, $\|Ax\|_\infty$.

设 $x = (x_1, x_2, \dots, x_n)$ 是 n 维向量, 则常用的向量范数有:

$$\|x\|_1 = \sum_{i=1}^n |x_i|$$

$$\|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$$

$$\|x\|_\infty = \max_{i=1}^n |x_i|$$

一般地, 定义 p 范数为:

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$$

设矩阵 A 是 $n \times n$ 的矩阵, 则常用的矩阵范数有:

$$\|A\|_1 = \max_{1 \leq j \leq n} \left\{ \sum_{i=1}^n |a_{ij}| \right\} \quad (\text{列范数})$$

$$\|A\|_F = \sqrt{\sum_{i=1}^n \sum_{j=1}^n |a_{ij}|^2} \quad (\text{Frobenius 范数})$$

$$\|A\|_\infty = \max_{1 \leq i \leq n} \left\{ \sum_{j=1}^n |a_{ij}| \right\} \quad (\text{行范数})$$

$$\|A\|_1 = \max_{1 \leq j \leq n} \left\{ \sum_{i=1}^n |a_{ij}| \right\} = \max\{3, 6\} = 6$$

$$\|A\|_2 = \sqrt{\sum_{i=1}^n \sum_{j=1}^n |a_{ij}|^2} = \sqrt{1^2 + 1^2 + 2^2 + 5^2} = \sqrt{31} \approx 5.77$$

$$\|A\|_\infty = \max_{1 \leq i \leq n} \left\{ \sum_{j=1}^n |a_{ij}| \right\} = \max\{2, 7\} = 7$$

$$Ax = \begin{pmatrix} 1 & 1 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 19 \end{pmatrix}$$

$$\|Ax\|_1 = \sum_{i=1}^n |Ax_i| = 5 + 19 = 24$$

$$\|Ax\|_2 = \sqrt{\sum_{i=1}^n |Ax_i|^2} = \sqrt{5^2 + 19^2} = \sqrt{386} \approx 19.65$$

$$\|Ax\|_\infty = \max_{1 \leq i \leq n} |Ax_i| = \max\{5, 19\} = 19$$

2. 设 $A = \begin{pmatrix} 2 & 1 \\ -5 & 4 \end{pmatrix}$, 则 A 的谱半径 $\rho(A)$ 和条件数 $\text{Cond}(A)_\infty$ 怎样计算?

谱半径 $\rho(A)$ 是 A 的所有特征值的绝对值/模的最大值。

设 A^{-1} 存在, 称数 $\|A\|\|A^{-1}\|$ 为 A 的条件数, 记为 $\text{Cond}(A)$ 。

常用的条件数如下:

(1)

$$\text{Cond}(A)_{\infty} = \|A\|_{\infty} \cdot \|A^{-1}\|_{\infty}$$

(2)

$$\text{Cond}(A)_2 = \|A\|_2 \cdot \|A^{-1}\|_2 = \sqrt{\frac{\lambda_{\max}(A^T A)}{\lambda_{\min}(A^T A)}}$$

$$|A - \lambda E| = \begin{vmatrix} 2 - \lambda & 1 \\ -5 & 4 - \lambda \end{vmatrix} = (2 - \lambda)(4 - \lambda) + 5 = \lambda^2 - 6\lambda + 13 = 0$$

$$\lambda = \frac{6 \pm \sqrt{36 - 52}}{2} = 3 \pm 2i$$

$$\rho(A) = \max |\lambda| = \sqrt{3^2 + 2^2} = \sqrt{13}$$

$$|A| = 2 \times 4 - 1 \times (-5) = 13$$

$$A^{-1} = \frac{1}{13} \begin{pmatrix} 4 & -1 \\ 5 & 2 \end{pmatrix}$$

$$\|A\|_{\infty} = \max(2 + 1, 5 + 4) = 9$$

$$\|A^{-1}\|_{\infty} = \frac{1}{13} \max(4 + 1, 5 + 2) = \frac{7}{13}$$

$$\text{Cond}(A)_{\infty} = \|A\|_{\infty} \cdot \|A^{-1}\|_{\infty} = 9 \times \frac{7}{13} = \frac{63}{13}$$

18. 某一装置运动轨迹为一圆锥曲线 $x^2 + bxy + cy^2 + dx + ey + f = 0$, 在运动轨迹上测得 5 个不同的点: $c_1(14.38, 3.94)$, $c_2(11.38, 2.79)$, $c_3(7.42, 3.07)$, $c_4(6.38, 5.11)$, $c_5(8.81, 2.59)$, 试形成 b, c, d, e, f 所满足的方程, 并用列主元消去法求出 b, c, d, e, f 的近似值。

列主元消去法, 其实就是在我们线性代数中的高斯消元法的基础上, 每一步消元之前在当前列中找到绝对值最大的行, 交换当前行和最大行, 再进行消元。高斯消元法, 就是通过矩阵的行变换, 将矩阵 A 变成上三角矩阵 U , 再通过 U 的解, 得到 $Ax = b$ 的解。显然, 在这道题中, 计算量将会非常大, 所以我们只需掌握思路即可。

将各点代入 $x^2 + bxy + cy^2 + dx + ey + f = 0$, 整理得:

$$\begin{bmatrix} x_1 y_1 & y_1^2 & x_1 & y_1 & 1 \\ x_2 y_2 & y_2^2 & x_2 & y_2 & 1 \\ x_3 y_3 & y_3^2 & x_3 & y_3 & 1 \\ x_4 y_4 & y_4^2 & x_4 & y_4 & 1 \\ x_5 y_5 & y_5^2 & x_5 & y_5 & 1 \end{bmatrix} \begin{bmatrix} b \\ c \\ d \\ e \\ f \end{bmatrix} = \begin{bmatrix} -x_1^2 \\ -x_2^2 \\ -x_3^2 \\ -x_4^2 \\ -x_5^2 \end{bmatrix}$$

代入数据:

$$\begin{bmatrix} 56.6572 & 15.5236 & 14.38 & 3.94 & 1 \\ 31.7502 & 7.7841 & 11.38 & 2.79 & 1 \\ 22.7794 & 9.4249 & 7.42 & 3.07 & 1 \\ 32.6018 & 26.1121 & 6.38 & 5.11 & 1 \\ 22.8179 & 6.7081 & 8.81 & 2.59 & 1 \end{bmatrix} \begin{bmatrix} b \\ c \\ d \\ e \\ f \end{bmatrix} = \begin{bmatrix} -206.7844 \\ -129.5044 \\ -55.0564 \\ -40.7044 \\ -77.6161 \end{bmatrix}$$

构造增广矩阵，进行列主元消去：

$$\left[\begin{array}{ccccc|c} 56.6572 & 15.5236 & 14.3800 & 3.9400 & 1.0000 & -206.7844 \\ 31.7502 & 7.7841 & 11.3800 & 2.7900 & 1.0000 & -129.5044 \\ 22.7794 & 9.4249 & 7.4200 & 3.0700 & 1.0000 & -55.0564 \\ 32.6018 & 26.1121 & 6.3800 & 5.1100 & 1.0000 & -40.7044 \\ 22.8179 & 6.7081 & 8.8100 & 2.5900 & 1.0000 & -77.6161 \end{array} \right]$$

$$R_2 - 0.5604 \times R_1 \rightarrow R_2:$$

$$\left[\begin{array}{ccccc|c} 56.6572 & 15.5236 & 14.3800 & 3.9400 & 1.0000 & -206.7844 \\ 0.0000 & -0.9152 & 3.3216 & 0.5821 & 0.4396 & -13.6242 \\ 22.7794 & 9.4249 & 7.4200 & 3.0700 & 1.0000 & -55.0564 \\ 32.6018 & 26.1121 & 6.3800 & 5.1100 & 1.0000 & -40.7044 \\ 22.8179 & 6.7081 & 8.8100 & 2.5900 & 1.0000 & -77.6161 \end{array} \right]$$

$$R_3 - 0.4021 \times R_1 \rightarrow R_3:$$

$$\left[\begin{array}{ccccc|c} 56.6572 & 15.5236 & 14.3800 & 3.9400 & 1.0000 & -206.7844 \\ 0.0000 & -0.9152 & 3.3216 & 0.5821 & 0.4396 & -13.6242 \\ 0.0000 & 3.1835 & 1.6384 & 1.4859 & 0.5979 & 28.0826 \\ 32.6018 & 26.1121 & 6.3800 & 5.1100 & 1.0000 & -40.7044 \\ 22.8179 & 6.7081 & 8.8100 & 2.5900 & 1.0000 & -77.6161 \end{array} \right]$$

$$R_4 - 0.5754 \times R_1 \rightarrow R_4:$$

$$\left[\begin{array}{ccccc|c} 56.6572 & 15.5236 & 14.3800 & 3.9400 & 1.0000 & -206.7844 \\ 0.0000 & -0.9152 & 3.3216 & 0.5821 & 0.4396 & -13.6242 \\ 0.0000 & 3.1835 & 1.6384 & 1.4859 & 0.5979 & 28.0826 \\ 0.0000 & 17.1795 & -1.8946 & 2.8428 & 0.4246 & 78.2839 \\ 22.8179 & 6.7081 & 8.8100 & 2.5900 & 1.0000 & -77.6161 \end{array} \right]$$

$$R_5 - 0.4027 \times R_1 \rightarrow R_5:$$

$$\left[\begin{array}{ccccc|c} 56.6572 & 15.5236 & 14.3800 & 3.9400 & 1.0000 & -206.7844 \\ 0.0000 & -0.9152 & 3.3216 & 0.5821 & 0.4396 & -13.6242 \\ 0.0000 & 3.1835 & 1.6384 & 1.4859 & 0.5979 & 28.0826 \\ 0.0000 & 17.1795 & -1.8946 & 2.8428 & 0.4246 & 78.2839 \\ 0.0000 & 0.4562 & 3.0187 & 1.0032 & 0.5973 & 5.6634 \end{array} \right]$$

第 2 列主元在第 4 行，交换第 2 行与第 4 行：

$$\left[\begin{array}{ccccc|c} 56.6572 & 15.5236 & 14.3800 & 3.9400 & 1.0000 & -206.7844 \\ 0.0000 & 17.1795 & -1.8946 & 2.8428 & 0.4246 & 78.2839 \\ 0.0000 & 3.1835 & 1.6384 & 1.4859 & 0.5979 & 28.0826 \\ 0.0000 & -0.9152 & 3.3216 & 0.5821 & 0.4396 & -13.6242 \\ 0.0000 & 0.4562 & 3.0187 & 1.0032 & 0.5973 & 5.6634 \end{array} \right]$$

$$R_3 - 0.1853 \times R_2 \rightarrow R_3:$$

$$\left[\begin{array}{ccccc|c} 56.6572 & 15.5236 & 14.3800 & 3.9400 & 1.0000 & -206.7844 \\ 0.0000 & 17.1795 & -1.8946 & 2.8428 & 0.4246 & 78.2839 \\ 0.0000 & 0.0000 & 1.9895 & 0.9591 & 0.5193 & 13.5758 \\ 0.0000 & -0.9152 & 3.3216 & 0.5821 & 0.4396 & -13.6242 \\ 0.0000 & 0.4562 & 3.0187 & 1.0032 & 0.5973 & 5.6634 \end{array} \right]$$

$$R_4 - (-0.0533) \times R_2 \rightarrow R_4:$$

$$\left[\begin{array}{ccccc|c} 56.6572 & 15.5236 & 14.3800 & 3.9400 & 1.0000 & -206.7844 \\ 0.0000 & 17.1795 & -1.8946 & 2.8428 & 0.4246 & 78.2839 \\ 0.0000 & 0.0000 & 1.9895 & 0.9591 & 0.5193 & 13.5758 \\ 0.0000 & 0.0000 & 3.2206 & 0.7335 & 0.4622 & -9.4539 \\ 0.0000 & 0.4562 & 3.0187 & 1.0032 & 0.5973 & 5.6634 \end{array} \right]$$

$$R_5 - 0.0266 \times R_2 \rightarrow R_5:$$

$$\left[\begin{array}{ccccc|c} 56.6572 & 15.5236 & 14.3800 & 3.9400 & 1.0000 & -206.7844 \\ 0.0000 & 17.1795 & -1.8946 & 2.8428 & 0.4246 & 78.2839 \\ 0.0000 & 0.0000 & 1.9895 & 0.9591 & 0.5193 & 13.5758 \\ 0.0000 & 0.0000 & 3.2206 & 0.7335 & 0.4622 & -9.4539 \\ 0.0000 & 0.0000 & 3.0690 & 0.9277 & 0.5860 & 3.5847 \end{array} \right]$$

第 3 列主元在第 4 行，交换第 3 行与第 4 行：

$$\left[\begin{array}{ccccc|c} 56.6572 & 15.5236 & 14.3800 & 3.9400 & 1.0000 & -206.7844 \\ 0.0000 & 17.1795 & -1.8946 & 2.8428 & 0.4246 & 78.2839 \\ 0.0000 & 0.0000 & 3.2206 & 0.7335 & 0.4622 & -9.4539 \\ 0.0000 & 0.0000 & 1.9895 & 0.9591 & 0.5193 & 13.5758 \\ 0.0000 & 0.0000 & 3.0690 & 0.9277 & 0.5860 & 3.5847 \end{array} \right]$$

$$R_4 - 0.6177 \times R_3 \rightarrow R_4:$$

$$\left[\begin{array}{ccccc|c} 56.6572 & 15.5236 & 14.3800 & 3.9400 & 1.0000 & -206.7844 \\ 0.0000 & 17.1795 & -1.8946 & 2.8428 & 0.4246 & 78.2839 \\ 0.0000 & 0.0000 & 3.2206 & 0.7335 & 0.4622 & -9.4539 \\ 0.0000 & 0.0000 & 0.0000 & 0.5060 & 0.2337 & 19.4158 \\ 0.0000 & 0.0000 & 3.0690 & 0.9277 & 0.5860 & 3.5847 \end{array} \right]$$

$$R_5 - 0.9529 \times R_3 \rightarrow R_5:$$

$$\left[\begin{array}{ccccc|c} 56.6572 & 15.5236 & 14.3800 & 3.9400 & 1.0000 & -206.7844 \\ 0.0000 & 17.1795 & -1.8946 & 2.8428 & 0.4246 & 78.2839 \\ 0.0000 & 0.0000 & 3.2206 & 0.7335 & 0.4622 & -9.4539 \\ 0.0000 & 0.0000 & 0.0000 & 0.5060 & 0.2337 & 19.4158 \\ 0.0000 & 0.0000 & 0.0000 & 0.2288 & 0.1455 & 12.5933 \end{array} \right]$$

$$R_5 - 0.4521 \times R_4 \rightarrow R_5:$$

$$\left[\begin{array}{ccccc|c} 56.6572 & 15.5236 & 14.3800 & 3.9400 & 1.0000 & -206.7844 \\ 0.0000 & 17.1795 & -1.8946 & 2.8428 & 0.4246 & 78.2839 \\ 0.0000 & 0.0000 & 3.2206 & 0.7335 & 0.4622 & -9.4539 \\ 0.0000 & 0.0000 & 0.0000 & 0.5060 & 0.2337 & 19.4158 \\ 0.0000 & 0.0000 & 0.0000 & 0.0000 & 0.0399 & 3.8146 \end{array} \right]$$

回代求解：

$$f = \frac{3.8146}{0.0399} \approx 95.7173$$

$$e = \frac{19.4158 - 0.2337 \times 95.7173}{0.5060} \approx -5.8426$$

$$d = \frac{-9.4539 - 0.7335 \times (-5.8426) - 0.4622 \times 95.7173}{3.2206} \approx -15.3421$$

$$c = \frac{78.2839 - (-1.8946) \times (-15.3421) - 2.8428 \times (-5.8426) - 0.4246 \times 95.7173}{17.1795} \approx 1.4661$$

$$b = \frac{-206.7844 - 15.5236 \times 1.4661 - 14.3800 \times (-15.3421) - 3.9400 \times (-5.8426) - 1.0000 \times 95.7173}{56.6572} \approx -1.4406$$

解得：

$$b \approx -1.4406, \quad c \approx 1.4661, \quad d \approx -15.3421, \quad e \approx -5.8426, \quad f \approx 95.7173$$